



Mourya Sai <mourya1706@gmail.com>

Request for Guidance on Reproducing and Extending Your MultiSSE Research

4 messages

Mourya Sai <mourya1706@gmail.com>

17 September 2026 at 15:10

To: entrup@sra.uni-hannover.de

Cc: kaessens@sra.uni-hannover.de, fiedler@sra.uni-hannover.de, lohmann@sra.uni-hannover.de

Dear Dr. Entrup and the Authors,

I am P. Mourya Sai, a final-year B.E. student from Vasavi College of Engineering, Hyderabad, India. My team and I are currently working on a research project based on your paper, "Applied static analysis and specialization of cross-core syscalls for multi-core AUTOSAR OS."

We are attempting to reproduce and validate the MultiSSE approach presented in your work as the first phase of our project. We would be very grateful if you could provide us with some guidance regarding the practical implementation and evaluation of the approach.

In particular, we would appreciate any advice on the following areas:

1. Reproduction of MultiSSE: What would you recommend as the best starting point for reproducing the analysis and optimization pipeline described in the paper?
2. ARA and LLVM: Are there any specific versions, repositories, branches, commits, or configurations of ARA and LLVM that you recommend using to reproduce the results?
3. AUTOSAR/RTOS environment: We are trying to understand how to set up the required AUTOSAR-compatible environment using tools such as dOSEK and Trampoline. Are there any particular versions or configurations that would be most suitable?
4. Evaluation methodology: What would you consider the minimum set of experiments or benchmarks that should be reproduced to demonstrate that our implementation is behaving consistently with the methodology presented in the paper?
5. Timing and performance measurements: We plan to use a Raspberry Pi 5 for our experimental platform. Since the paper evaluates the approach on a different hardware platform, are there any important considerations or adjustments we should make when comparing our measurements?
6. Common implementation difficulties: Are there any parts of the MultiSSE implementation that you found particularly difficult or that we should pay special attention to when attempting a reproduction?
7. Further extension: After reproducing the MultiSSE approach, our project aims to investigate the incorporation of mixed-signal timing information from automotive sensor interfaces into AUTOSAR timing analysis. If you have any recommendations regarding how such an extension could be connected to the existing MultiSSE framework, we would greatly appreciate your suggestions.

We understand that you may be busy, so even a few pointers to relevant repositories, documentation, previous work, or recommended starting points would be extremely helpful to us.

Thank you for your time and for making this work available to the research community.

Kind regards,

P. Mourya Sai
Final-year B.E. Student
Vasavi College of Engineering
Hyderabad, India

Andreas Kässens <kaessens@sra.uni-hannover.de>

17 September 2026 at 17:52

To: Mourya Sai <mourya1706@gmail.com>

Cc: fiedler@sra.uni-hannover.de, lohmann@sra.uni-hannover.de, entrup@sra.uni-hannover.de, Mareike Burg <burg@sra.uni-hannover.de>

Hello P. Mourya Sai,

Thank you for your interest in our work on MultiSSE and I'm happy to answer your questions:

1+2. You can find the original artifact for MultiSSE (without synthesis) here [1].

The source code is available on GitHub [2] if you want to setup your own development environment, but this can be a hassle because of the requirement versions.

Alternatively, you can also use the updated and reproducible artifact for our follow-up paper HyperSSE [3]. This docker container also contains the previous MultiSSE, and I would recommend this one as it is more up to date.

3. The versioned requirements are included as meson subprojects in the PARROT meta-repository and automatically checked out to the right state. In the artifacts, the source code is already checked out.

4. We have several meson test suites, one specific for the MultiSSE analysis and one for the generator `autosar_generator_pi4`.

5. To evaluate on Raspberry Pi 5 you'll need to adapt the AUTOSAR libraries, including boot code and peripheral control, as they currently only support the Pi 4. This might take some time, using the Pi 4 would be easier.

6. In the loaded config json you have to explicitly enable timing annotations for the applications if you want them included.

```
"MultiSSE": {"with_times": true}}
```

With timing annotations, the analysis is more complex, but leads to better results (sparser state graph).

7. Our timing annotations are based on (atomic) basic blocks. You can provide a separate json file, e.g. 'timing': 'a_timings.json' or annotate them directly in the source code using a helper function 'ara_timing_info(a,b)' and enable them with 'timing': 'BB'.

Your timing information can be included in the same way.

Some important files you can look into first:

`parrot/subprojects/ara/appl/AUTOSAR/meson.build`

`parrot/subprojects/ara/settings/autosar_generator_arm.json`

`parrot/subprojects/ara/ara/steps/multisse.py`

I hope this serves you well as a starting point for your research project. If you find any improvements or develop extensions, we're also happy to take PRs via GitHub.

Best regards,
Andreas

[1] <https://www.sra.uni-hannover.de/Publications/2023/multisse-rtas23/>

[2] <https://github.com/luhsra/parrot>

[3] <https://drops.dagstuhl.de/entities/document/10.4230/DARTS.12.2.2>

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Mourya Sai <mourya1706@gmail.com>

22 September 2026 at 22:42

To: Andreas Kässens <kaessens@sra.uni-hannover.de>

Cc: fiedler@sra.uni-hannover.de, lohmann@sra.uni-hannover.de, entrup@sra.uni-hannover.de, Mareike Burg <burg@sra.uni-hannover.de>

Dear Andreas,

Thank you very much for your detailed response and for pointing us toward the HyperSSE artifact. This is extremely helpful and provides us with a great starting point for our project.

We appreciate the guidance on the implementation and the specific files to focus on. We will begin exploring these resources and look forward to working with the framework.

Thank you again for your time and support.

Best regards,

P. Mourya Sai

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Mourya Sai <mourya1706@gmail.com>

22 September 2026 at 22:42

To: harshavardhanedam@gmail.com

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